Putting Quantum AI Hype to the Test in Cislunar Space: A Governed Public-Data Benchmark for Human-Centered Trajectory-Correction Planning

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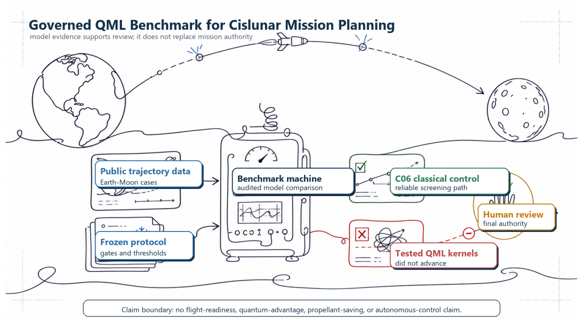


Fig. 1. Simulated one-line illustration: public trajectory data and a frozen protocol are assessed against a physics-residual control and QML kernels; final authority remains with human review.

Abstract. This study evaluates simulated quantum machine-learning (QML) kernels for a public-data, crewed Earth–Moon trajectory correction benchmark. The target is robust total correction delta-v, or modeled velocity change under uncertainty. Related cases shared folds, preprocessing used training rows only, thresholds were fixed before outcomes, and calibration and final-test rows remained locked. The simulator passed 67 eligible checks; one source-ineligible event was neither pass nor failure. Q01 produced mean normalized root-mean-square error (NRMSE) 0.646614 versus 0.008739 for the physics-residual classical control C06; lower is better. Q01b produced 0.661207 versus 0.006833 for C06. The feasibility-only kernel (FQK) retained 10.9% of actually feasible cases at the fixed 0.5 threshold. Sixteen bounded successors missed their predeclared advancement criteria, and the strongest raw improvement failed its matched classical control. Thus, the tested representations did not justify quantum complexity for this frozen task; this is not a universal claim about QML. The contribution is a transparent safeguard for high-consequence AI: provenance, negative-result reporting, and separation of crew requirements and human authorization from model scores. The study claims no quantum advantage, propellant saving, flight readiness, hardware performance, or QML invention.

Keywords: Quantum machine learning · human-centered AI · decision support · cislunar trajectory design · reproducible benchmarking · negative results

1 Introduction

Cislunar space is the region in and around the Earth–Moon system. A trajectory-correction maneuver, or correction burn, deliberately changes a spacecraft's velocity to keep its path within mission constraints. The required change is summarized by delta-v (measured in m/s). Delta-v is not propellant mass, although it is a standard measure of maneuver effort and is related to fuel use through the vehicle and propulsion model.

For a crew-carrying mission, correction planning is a coupled decision problem: burn timing and delta-v interact with navigation uncertainty, maneuver execution, crew timelines, entry conditions, and reserve constraints. Public Artemis trajectory-design research therefore treats robust correction cost, meaning correction delta-v evaluated with modeled uncertainty, together with mission constraints rather than as an unconstrained curve-fitting task [22]. A learning *surrogate*, a fast predictive model that approximates a more expensive simulation, could help screen candidate scenarios. That possibility is a motivation for testing, not evidence that a particular machine-learning model is useful or safe.

The paper adopts a human-centered decision-support framing. A model score is not the decision: a responsible reviewer needs to know which source data, physical assumptions, uncertainty boundary, and failure cases produced the score before deciding whether a model warrants further study. Crew-related requirements therefore remain separate from predictive performance, consistent with the intended-use and human-system boundaries in the applicable U.S. National Aeronautics and Space Administration (NASA) standards [8,9]. This work does not conduct a user study or measure usability, trust, workload, or operator performance. From a human–computer interaction (HCI) perspective, its contribution is the evidence design around a possible future decision-support tool.

Quantum machine learning (QML) uses quantum-state representations or quantum circuits inside a learning method. In a quantum-kernel model, a circuit maps each classical input to a quantum state and the overlap between two states is used as a similarity score. This offers feature maps different from common classical models [16,5], but difference alone does not imply usefulness. Encoding data introduces loading choices, similarity geometry, and resource costs. A circuit may be difficult to simulate in general while still being approximated well by a classical learner on the tested data [6,21]. Conversely, a favorable result can arise from a weakly tuned comparator, related mission cases leaking across a data split, thresholds chosen after seeing outcomes, or omitted failures. These are benchmarking problems before they are quantum-computing problems.

1.1 Problem statement

QML feature spaces can be mathematically distinct from classical feature spaces, but that distinction is not itself evidence of useful prediction. The literature has identified recurring risks of optimistic conclusions from weak classical comparators, information leakage across related examples, post-outcome model selection, or unreported negative outcomes [1,15]. In a crewed-spaceflight context, these risks create a knowledge gap: without a transparent, frozen comparison, a complex surrogate can appear promising before it has earned a bounded scientific claim. The problem addressed here is therefore both predictive and evidential—how to evaluate a QML surrogate fairly without allowing the evaluation process to manufacture a favorable conclusion.

1.2 Research objective

This paper documents a public-data experimental program for correction-cost prediction and feasibility screening. Its primary objective is to establish a governed, source-bound benchmark that evaluates QML representations against physics-based and matched classical controls under a frozen development protocol. The purpose was not to assert a priori that QML would win. It was to define a bounded task, verify the simulator against eligible public endpoints, preserve the data-access boundary, compare models fairly, retain negative outcomes, and prohibit claims that the evidence cannot support. This stance is consistent with model and simulation credibility practice [8] and with recent calls for fair QML comparisons [1,15].

1.3 Central research question

The study addresses one central research question:

> *To what extent can simulated quantum-kernel representations add empirical predictive value to a human-centered cislunar trajectory-correction decision-support benchmark when compared fairly with physics-based controls?*

This central question is resolved through four pre-specified operational questions, not four independent study aims:

1. Can the primary fidelity-style quantum kernel (repository identifier Q01) improve correction-cost prediction over physics-derived and classical controls under the frozen development protocol?
2. If Q01 fails, can the projected follow-up Q01b improve the same task without changing data access or data splits?
3. Can the feasibility-only quantum kernel (FQK) meet its fixed screening criteria for identifying independently propagated feasible cases?
4. Which conclusions remain valid when calibration, final-test, mission-loop, and hardware evidence stay locked?

The first three sub-questions received negative answers for the tested models and settings. The fourth establishes the boundary of that conclusion. Together, they yield a transparent evidence hierarchy that distinguishes simulator credibility, development-only model evidence, future-design lessons, and prohibited operational claims. Negative results are not treated as failed record keeping; they are the result when a hypothesis does not meet its predeclared standard.

The paper makes six contributions:

1. A source-bound cislunar benchmark protocol, meaning that inputs are tied to identified public files and recorded source versions, with grouped development folds, training-fold-only transformations, and retained numerical feasibility outcomes.
2. A simulator credibility chain that reports its eligible scope and its source-ineligible exception rather than converting missing evidence into a pass or failure.
3. A preregistered QML comparison against physics-residual, random-feature, compressed-input, and classical kernel controls.
4. A closed successor ledger containing exploratory and prospectively bounded negative candidates, including classical controls that replace the quantum component while keeping the surrounding construction matched.
5. A self-contained evidence and figure package that supports review of the negative conclusion without widening the study's scientific claim.
6. A human-centered claim boundary that keeps model evidence, crew-related requirements, and future human authorization distinct.

2 Scientific Scope and Claim Boundary

The study is an offline, public-data, development-only benchmark. It is not a mission design authority, a flight certification exercise, or a performance assessment of a quantum processor. A simulation result may be useful for evaluating a model family, but it cannot by itself establish the performance of a mission-owned trajectory system or a hardware quantum implementation. The distinction is essential for a human-rated application, where human-system requirements are separate from a surrogate's regression score [9].

2.1 Reader guide to labels and data stages

Short identifiers are retained because they link the paper to immutable repository records. They are audit keys, not scientific concepts that a reader is expected to know in advance:

Gate. An authorization checkpoint: Gate 3 checks eligible simulator scope, Gate 5 evaluates development models, and Gate 6 was never opened.

Decision and protocol IDs. D marks a governance decision or bounded campaign, P a prospective protocol; D046 is one successor-test record.

Model and figure IDs. C, Q, and A identify classical, QML, and analysis controls; C06 is the physics-residual control and Q01 the primary kernel. RFIG is a figure-provenance key, not a statistical quantity.

Development-only. Rows may select models; calibration would adjust a frozen output and final-test rows would estimate performance after choices are fixed. Neither was read by the reported campaigns.

Fold, seed, and rung. A fold preserves related groups, a seed fixes a reproducible random start, and a rung is a 128–1,024-row sample stage.

2.2 Human-centered decision-support stance

The model's role is offline screening information for qualified human review, not autonomous control, an operational recommendation, or a replacement for mission analysis. Its HCI contribution is evidence design: traceable scope, visible uncertainty and failures, and a fallback to physical analysis. No participants, interfaces, or measures of trust, workload, situation awareness, or decision quality were collected; the paper makes no empirical HCI claim.

The evidence boundary was defined before reporting. Gate 3 supports eligible simulator comparisons only; Gate 5 supports a frozen development decision; P001 and D034–D049 are development-only follow-ups that cannot revise Gate 5 or authorize Gate 6. The recall-first classical analysis is a future protocol-design lesson, not a safety filter. Calibration and final-test rows, mission-loop scenarios, quantum-hardware or graphics-processor execution, operational threshold use, trained-model release, and Gate 6 remain outside the study. Accordingly, the paper claims no quantum advantage, propellant saving, mission-loop validity, flight readiness, NASA approval, hardware behavior, final-test result, or QML invention.

Figure 2 records authorization state, not a model metric. A later candidate may be informative but does not become preregistered evidence merely because it is reported after the primary negative result.

3 Related Work and Positioning

The target problem is positioned relative to public work on Artemis trajectory-correction design, rather than being presented as a replacement for mission-owned analysis. Woffinden et al. describe a correction-burn placement formulation that incorporates uncertainty, crew-related timing, and entry constraints [22]. Classical learning research has explored low-thrust guidance and cislunar transfer design, but those settings differ in dynamics, propulsion, objectives, and validation structure [24,20,7,19,23,25]. They are valuable precedents for surrogate modeling, low-fidelity references, cost–reachability endpoints, and retaining infeasible cases, not interchangeable baselines for a human-rated correction benchmark.

Quantum feature-space methods encode a classical input as a quantum state and compare encoded states through a quantum-derived similarity, called a kernel, [16,5]. The method can be used in a kernel learner, but its predictive value

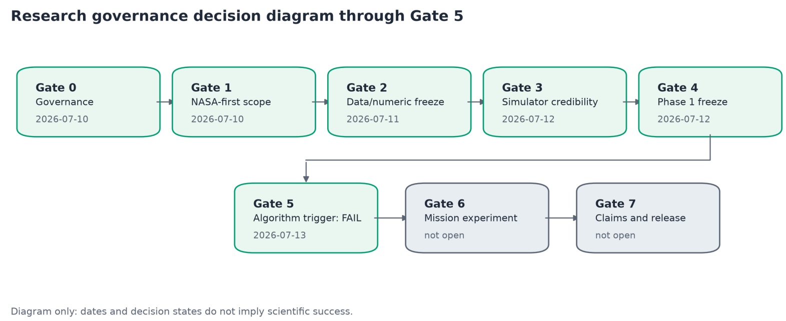


Fig. 2. Research governance through Gate 5 (RFIG-001). Gates 0–5 were closed by human decisions; the technical Gate 5 result is a benchmark-specific protocol FAIL, meaning its advancement hypothesis was rejected. Gate 6 remains unopened. This is decision-state evidence, not a model performance plot.

depends on whether that similarity reflects differences that matter for the target being predicted. Theoretical and numerical work has made this caveat increasingly explicit: hard-to-simulate circuits do not automatically give a learning advantage, and the data distribution can make a classical approximation competitive [6,21].

The present work is particularly informed by three benchmarking lessons. First, kernel bandwidth and encoding scale change the effective inductive bias, meaning the patterns that a model is structurally predisposed to represent; they are not negligible implementation details [17]. Second, a quantum-kernel study should include appropriately strong classical counterparts, including random features that approximate a distance-based classical kernel where relevant [10,14,21]. Third, a comparison should report more than its best run: model choice, split structure, seed variation, capacity, and post-outcome choices all affect what a claimed gain means [1,15,12].

3.1 Literature provenance of the frozen protocol

The literature set was used to shape the protocol, not to claim that every reviewed model was executed. Data re-uploading, which repeatedly inserts input features between circuit layers, encoding-dependent generalization, and initialization studies informed the need to freeze encoding, observables, initialization, and seed handling [11,2,4]. Quantum-agent work informed the need for matched classical capacity and explicit readout choices, but it did not turn this supervised benchmark into a reinforcement-learning study [18]. Quantum annealing for trajectory transcription was retained as adjacent quantum-optimization context and deliberately excluded from the QML candidate set [3].

The accompanying literature-source audit accounts for all 23 accepted records used in the experiment's review package and additionally identifies the random-Fourier source behind the A01 control. It distinguishes direct methodological influence from an empirical comparison in this paper. This prevents a citation from being mistaken for evidence that a reviewed external method was reproduced, validated, or outperformed here.

This paper does not make an exhaustive literature-review claim. The repository preserves a bounded extraction set of 23 accepted records used to establish the initial protocol and a separate archival discovery refresh. The refresh did not change the experiment, thresholds, figures, or outcomes. Literature is therefore used here to position the benchmark and justify controls, not to construct a retrospective theory of success.

4 Materials and Methods

4.1 Scenario construction and simulator credibility

The public pipeline froze three force-model levels: F0, central Earth point-mass gravity; F1, F0 plus Moon and Sun third-body gravity; and F2, F1 plus Earth J2 (Earth's equatorial-bulge correction) and tighter integration. Each retained the configured maneuver, mass-depletion, event, crew-constraint, uncertainty, and numerical-feasibility settings. The target is benchmark cost or feasibility, not spacecraft control; sources and scenario construction were frozen before model campaigns.

Gate 3 compared eligible Python endpoints (position and velocity at a window end) with GMAT under the same force-model configuration. The diagnostic chain isolated an Earth-J2 coefficient-file formatting error and confirmed its repair with degree-two, order-zero gravity comparisons. All 67 evaluable checks and the unchanged position and velocity limits passed. This supports only the tested public scope, not a complete mission environment.

RTC3 is `not_eligible`, not pass or fail: its qualified public trajectory file predates the event and cannot reconstruct its outcome under the frozen temporal-evidence rule. The visible exclusion distinguishes source eligibility from a numerical outcome.

4.2 Frozen development data and leakage control

Let $\mathcal{D} = \{(x_i, y_i, z_i, g_i)\}_{i=1}^{N}$ be the frozen development data: x_i contains allowed features, y_i is robust total correction delta-v in m/s, z_i is the independently propagated feasibility label, and g_i is the related mission-design group. Whole groups are assigned to one of five folds, so related cases cannot occur in both training and validation.

Imputation, category encoding, scaling, and configured principal-component analysis (PCA) are fit within each training fold only. Fitting them on all rows would leak held-out information. The deterministically selected 128, 256, 512, and 1,024-row rungs stop weak configurations under a predeclared rule; retained configurations use 20 frozen seeds. These are development stress tests, not operational-distribution estimates, and never read calibration or final-test data.

4.3 Task definitions and estimands

The primary endpoint is robust correction-cost prediction. For predictor f, normalized root-mean-square error (NRMSE) is

$$\mathrm{NRMSE}(f) = \frac{1}{s_y}\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(f(x_i) - y_i\right)^2}, \tag{1}$$

Here n is the held-out prediction count, $f(x_i)$ the predicted delta-v, y_i its simulated target, and s_y the frozen development target standard deviation. The numerator is root-mean-square error in m/s; division by s_y makes NRMSE unitless and comparable across folds. Zero is perfect and lower is better; 0.01 means error equal to 1% of the target standard deviation. The report pools out-of-fold predictions and retains fold/seed summaries. The feasibility-constrained regret is interpretive, not a replacement endpoint.

The classification track estimates $p_f(x) \in [0, 1]$, the probability that x is feasible under independent propagation. Its Brier score is

$$\mathrm{Brier} = \frac{1}{n}\sum_{i=1}^{n}\left(p_f(x_i) - z_i\right)^2, \tag{2}$$

where $z_i \in \{0, 1\}$ is the observed label; lower is better and zero is perfect. AUROC measures ranking across thresholds (0.5 chance; 1.0 perfect). At fixed $t = 0.5$, recall is

$$\mathrm{Recall}_{0.5} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{3}$$

TP and FN are correctly retained and incorrectly rejected feasible plans, so recall is the retained feasible fraction. Precision is the fraction of admitted plans that are feasible; an infeasible admission is a false positive. Thus low recall reduces utility, while unsafe acceptance requires precision, false-positive, and independent-propagation analysis. No one metric establishes operational safety.

4.4 Quantum-kernel representation

Q01 is an exact classical statevector simulation, not a quantum-processor result. For $q \in \{4, 6, 8\}$ qubits, its feature state is in $\mathcal{H}_q = (\mathbb{C}^2)^{\otimes q} \cong \mathbb{C}^{2^q}$ with the usual inner product. The implementation stores a pure statevector, so shots, hardware noise, and device-connectivity constraints do not affect the kernel.

For frozen numeric vectors $x, x' \in \mathbb{R}^d$, the deterministic map applies $U_\phi(x)$ to the all-zero state:

$$|\psi_\phi(x)\rangle = U_\phi(x)|0^{\otimes q}\rangle, \qquad \rho_\phi(x) = |\psi_\phi(x)\rangle\langle\psi_\phi(x)|. \tag{4}$$

Here ϕ is the fixed circuit/encoding configuration, $|\psi_\phi(x)\rangle$ its statevector, and $\rho_\phi(x)$ equivalent density-operator notation; computation uses the statevector directly. The feature dimension, encoding, and circuit are preserved in committed trials.

The corresponding fidelity-style kernel is

$$K_q(x, x') = \left|\left\langle 0^{\otimes q} \middle| U_\phi(x)^\dagger U_\phi(x') \middle| 0^{\otimes q} \right\rangle\right|^2, \tag{5}$$

For pure states, this equals $\mathrm{Tr}[\rho_\phi(x)\rho_\phi(x')]$ [16,5]; $\dagger$ is conjugate transpose. $K_q \in [0, 1]$ measures encoded-state similarity, not feasibility probability. Pairwise values form the Gram matrix used by a kernel regressor. Exact features, PCA dimensions, ϕ, bandwidths, and regularization are committed. Equation (5) defines a classically simulated representation, not hardware performance, speedup, or intrinsic task value.

4.5 Models and matched controls

Q01 is a fidelity-style correction-cost kernel regressor. Its main comparator is physics-residual C06, which adds learned correction $r(x)$ to a lower-fidelity physical estimate $b_{\mathrm{low}}(x)$:

$$\widehat{y}_{\mathrm{C06}}(x) = b_{\mathrm{low}}(x) + r(x). \tag{6}$$

Unlike a generic regressor, C06 begins with available physical structure and learns only residual error. Controls also include A01 random Fourier features, a finite classical approximation to a radial-basis-function (RBF) kernel, compressed neural views, and frozen alternatives. They test whether a classical representation explains an apparent quantum-feature effect [19,23,14].

After Gate 5, P001 defined two follow-ups without reopening Q01. Q01b replaces whole-state overlap with local measurement expectations and a distance kernel; a deterministic Nyström approximation uses at most 256 training landmarks. FQK means *feasibility-only quantum kernel* and applies the same projected construction to independently propagated feasibility labels. Both retain groupwise splits, training-fold-only preprocessing, nested rungs, controls, and 20 seeds; they create development-only evidence and cannot revise Q01.

Each D034–D049 protocol froze its candidate, target, comparator, threshold, stopping rule, and source path before fitting. Names are immutable registry keys. Lower NRMSE than C06 was insufficient: a candidate also had to beat a matched classical alternative with the same inputs and stages. This dequantization control asks whether benefit survives replacement of the quantum-derived component by a classical one.

4.6 Human-centered decision-support boundary

Outputs are evidence for a future screening tool, not trajectory commands. A reviewer needs the estimate, scenario group, comparator, uncertainty record, and authorized scope; the package preserves these rather than a leaderboard alone. Ineligible, poorly calibrated, source-ineligible, or out-of-scope candidates are not promoted and the physical/numerical reference remains the decision basis. This is an HCI design boundary, not an interface study.

4.7 Statistical decision process

The Gate 5 trigger required all eligible folds/seeds, the strongest classical finalist and matched dequantization controls, paired uncertainty checks, and at least one qualifying predeclared regime (for example, fidelity or uncertainty family). Paired errors use the same folds/seeds; 95% bootstrap intervals resample their differences, sign-permutation tests assess direction under no advantage, and multiplicity control limits repeated-comparison false positives. No post-outcome seed, model, split, or threshold was added. Learning curves and retained seeds guard against conclusions from training fit or one favorable run [13,4].

D006 completed 871 of 871 authorized tasks with zero software or task failures. Q02 and Q03 did not reach larger rungs because too few first-rung configurations met frozen retention; their status is `not_reached_under_frozen_eligibility`, not failed, missing, or withheld evidence.

4.8 Computational-resource and reproducibility boundary

The reference environment was a Windows laptop with a 13th Generation Intel Core i9-13900HX CPU, 32 GB RAM, and NVIDIA GeForce RTX 4060 Laptop GPU. A q-qubit statevector stores 2^q complex amplitudes; tasks ran one at a time within the memory bound. Neither GPU nor quantum hardware produced reported QML evidence. The specification explains throughput constraints, not a required future setup. The release supplies configurations, seeds, data cards, figure registry, tables, and scripts, preserving auditability and locked-row boundaries.

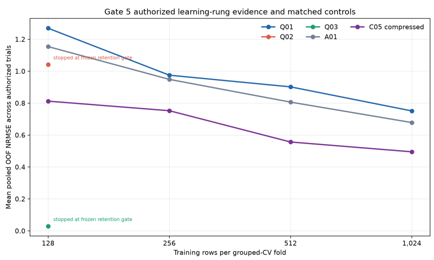
5 Results

5.1 Simulator credibility result

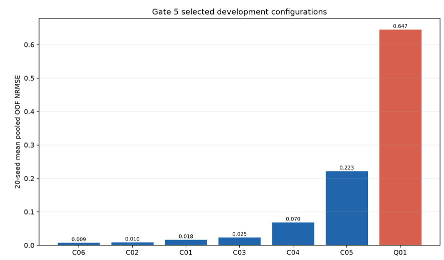
The accepted Gate 3 record contains 67 evaluable checks and zero failures. The result includes the Earth-only diagnostic chain that localized the original J2 discrepancy to a coefficient-file delimiter problem rather than to the integrator, time system, or initial state. Once the coefficient format was repaired, the GMAT–Python comparison passed the unchanged position and velocity thresholds across the eligible windows. The appropriate conclusion is that the public simulator implements the tested scope consistently enough to support the development benchmark. It is not a flight-readiness conclusion, and the RTC3 event remains untested because no eligible post-event public source was available. RTC3 is therefore neither a pass nor a failure.

5.2 Preregistered Gate 5 benchmark

The Q01 quantum kernel did not satisfy the development trigger. Its mean NRMSE was 0.646614. The physics-residual control C06 achieved 0.008739, making Q01's normalized error about 74 times as large. Because lower NRMSE is



(a) Learning curves



(b) Seed summary

No regime cell satisfied every preregistered condition.
Preregistered Gate 5 trigger: **FAIL**; calibration reads = 0;
final-test reads = 0.

(c) Algorithm-trigger controls

Fig. 3. Gate 5 development evidence. (a) Learning curves and matched controls (RFIG-021); (b) the 20-seed stability diagnostic (RFIG-022); and (c) preregistered trigger controls (RFIG-023). All panels are development-only; a passing trigger would have supported one further algorithm study, not quantum advantage or operational deployment.

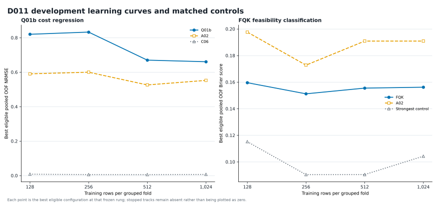
better, this was not a close comparison that depended on one uncertainty test. No predeclared subgroup satisfied the advance rule. Gate 5 is therefore a valid technical `FAIL` for the specified Q01 model, task, preprocessing, controls, and development population. Here `FAIL` is a protocol decision meaning that the performance hypothesis was rejected. It is not a software crash and not a claim that every possible QML method fails.

Figure 3 combines the pooled learning curves, the 20-seed selected-model distribution, and the preregistered trigger controls. The panels show that a model was assessed by its retained path through the precommitted rules, not by one aggregate mean.

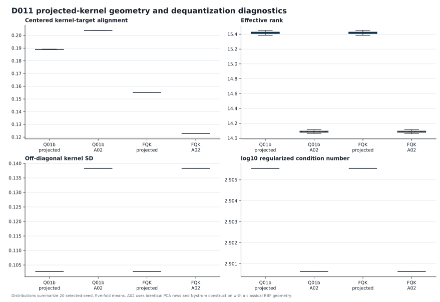
Q02 and Q03 require a precise reading. They completed their authorized tasks, but too few configurations of the variational quantum regressor (Q02) and hybrid quantum-residual model (Q03) satisfied the frozen eligibility condition for larger rungs and seed reruns. They are recorded as "not reached under frozen eligibility" (`not_reached_under_frozen_eligibility`). Treating this state as a failed experiment would overstate the evidence; treating it as an unreported potential success would ignore the protocol. The paper reports the actual decision state.

5.3 Post-Gate 5 exploratory tracks

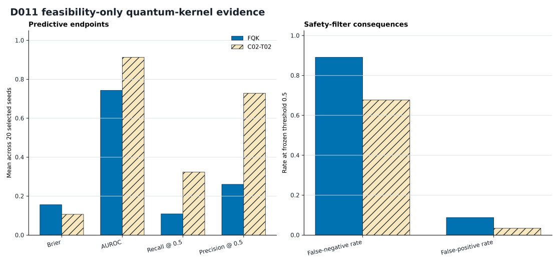
Q01b tested whether changing the quantum readout while keeping the same development discipline could materially alter the result. It did not. Q01b attained mean NRMSE 0.661207, compared with 0.006833 for C06, so its error was about 97 times as large. Figure 4, panels (a)–(b), shows the learning curves, projected-kernel geometry, and dequantization diagnostics. For this projected map, the



(a) Q01b learning curves



(b) Q01b geometry and dequantization



(c) FQK screening endpoints

Fig. 4. Post-Gate 5 development-only diagnostics. (a) Q01b learning curves and matched controls (D011/P001; RFIG-026); (b) Q01b geometry compared with the A02 classical RBF kernel on matched training-fold PCA rows and Nyström landmarks (RFIG-027); and (c) FQK Brier score, AUROC, recall, precision, and consequence metrics (RFIG-028). None revises the Gate 5 decision.

representation did not capture residual structure better than the classical controls; this does not rule out projected kernels on other physical problems.

The FQK classification track separates discrimination from a safety-filter interpretation. It achieved AUROC 0.7436 and Brier score 0.1561, but recall at the fixed threshold was only 0.1089. Because the positive label is "independently propagated feasible," this means FQK retained only about 10.9% of actually feasible plans and rejected the remaining 89.1%. That low utility prevents promotion to the proposed screening role. It does not by itself measure unsafe acceptance, which instead requires precision and false-positive analysis. Panel (c) of Fig. 4 keeps these meanings separate. FQK cannot be called useful for safety merely because one metric is above chance or because a quantum feature map was evaluated.

The recall-first classical feasibility analysis, tracked as CSAFE-RF in the repository, provides a separate lesson for a future protocol. A calibrated logistic model reached mean feasible-case recall 0.804253, false-negative rate 0.195747, and Brier score 0.142231. In plain terms, it retained about 80.4% of actually feasible plans and rejected about 19.6% of them. This exposes a tradeoff between calibrated probabilities and retaining useful options, but it does not establish safety: a future screening protocol must separately freeze a feasible-case

Table 1. Closed D034–D049 successor branch. Candidate names are immutable protocol IDs; their definitions are preserved in the repository. Every row is a valid development-only negative. The C06 NRMSE was 0.006833 throughout. Lower NRMSE is better, and a negative paired difference favors the candidate over C06, but each branch also had to pass its matched-control and stopping rules.

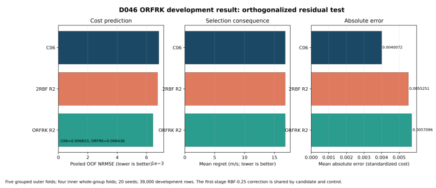
Decision	Candidate	Candidate NRMSE	Difference vs C06	95% paired interval
D034	PRQK-08-N	0.029326	0.022493	[0.022486, 0.022500]
D035	CFQSR-08-N	0.013463	0.006630	[0.006627, 0.006632]
D036	TAPQK-08	0.010382	0.003549	[0.003547, 0.003551]
D037	TSQR-08-L025	0.006848	0.000015	[0.000015, 0.000015]
D038	GFRK-08-L010	0.006647	-0.000186	[-0.000187, -0.000184]
D039	EC-GFRK-08-L010	0.006466	-0.000366	[-0.000368, -0.000365]
D040	CE-GFRK-08-L010	0.006504	-0.000329	[-0.000330, -0.000328]
D041	HEFRK-08-E050	0.006592	-0.000241	[-0.000242, -0.000240]
D042	AGEFRK-08-ADAPT	0.006524	-0.000309	[-0.000310, -0.000308]
D043	SFRK-08-CV	0.006445	-0.000387	[-0.000389, -0.000386]
D044	NIFRK-08-NL	0.006751	-0.000081	[-0.000084, -0.000079]
D045	MSFRK-ALL	0.006731	-0.000102	[-0.000103, -0.000100]
D046	ORFRK-08-R2	0.006436	-0.000397	[-0.000399, -0.000394]
D047	OMFRK-ALL	0.006952	0.000119	[0.000113, 0.000125]
D048	NSORFRK-08	0.006590	-0.000243	[-0.000244, -0.000242]
D049	SSORFRK-08	0.006566	-0.000266	[-0.000268, -0.000265]

recall floor, an infeasible-case false-positive limit, and a calibration requirement. CSAFE-RF was not used to re-rank QML candidates, change a threshold, or authorize a mission filter.

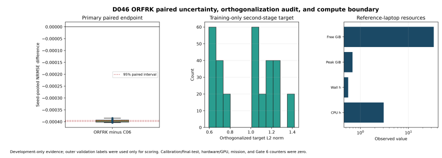
5.4 Closed successor branch D034–D049

The closed successor branch contains 16 prospectively bounded candidates. All read the development scope only, used 39,000 development rows per campaign, and completed with the recorded status `complete_valid_negative`: the computation and integrity checks succeeded, but the scientific advancement rule did not. The complete results are listed in Table 1. Candidate names in the table are traceability IDs for the frozen protocols, not terminology needed to interpret the conclusion. The table is intentionally included in full rather than selecting only candidates with lower raw error than C06.

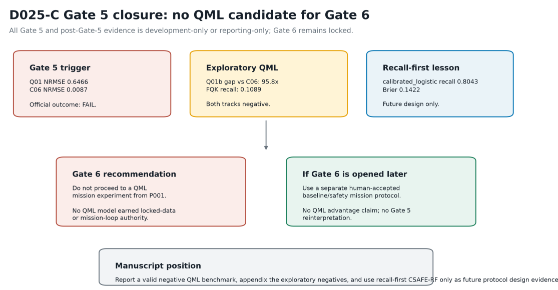
Several variants have a lower mean NRMSE than C06. This is a reason to inspect the controls, not a reason to announce a quantum-specific result. The best raw C06 comparison was decision D046's eight-qubit orthogonalized residual fidelity–RBF kernel (ORFRK-08-R2): NRMSE 0.0064363 versus 0.0068328 for C06, an apparent 5.80% reduction. Both the candidate and its matched classical control first removed residual structure captured by the same RBF stage. The candidate then used a quantum-fidelity channel, while the TWO-RBF control used a second classical RBF channel. TWO-RBF attained 0.0067501, so the candidate's improvement over that stronger like-for-like control was about 4.65%, below the 5% rule fixed before the run. The result therefore does not meet the evidence standard for a quantum-specific candidate.



(a) D046 result comparison



(b) Paired audit and resources



(c) Closure and Gate 6 recommendation

Fig. 5. Closure evidence. (a) D046 ORFRK result comparison (RFIG-078): the $q = 8$ statevector candidate and matched classical control share the first RBF correction and differ only in the second-stage channel. (b) Paired audit and resource boundary (RFIG-079), with no quantum-hardware or GPU result. (c) Gate 5 closure (RFIG-044): no closed-branch candidate is eligible for Gate 6.

5.5 Evidence accounting and closure

Evidence accounting confirms that the reported conclusion follows the authorized boundary. The preregistered development campaign (D006) completed 871 of 871 tasks, with zero task failures and zero reads of calibration or final-test data. The projected-kernel exploratory campaign (D011-R1) read 39,000 development rows and zero locked rows. No hardware quantum job, mission-loop run, Gate 6 run, or final-test evaluation occurred.

Figure 5 summarizes the D046 comparison, its paired audit, and the closure recommendation. No candidate is eligible for Gate 6; the figure does not represent an unexecuted Gate 6 experiment. Closure decision D055-C ended active experimentation so that future protocol ideas could not be mistaken for completed evidence during publication preparation.

6 Discussion

The principal finding is a controlled negative benchmark: the runs completed validly, but the tested performance hypotheses were rejected. The Q01 fidelity-style quantum kernel was far behind the physics-residual control, and the Q01b projected feature map did not close the gap. The feasibility-only kernel did not meet its fixed multi-metric screening criteria. The successor branch demonstrates an additional failure mode that matters beyond this task: a candidate can improve over one baseline yet fail against a classical model with the same staged construction. That distinction is why matched dequantization controls and thresholds fixed before outcomes belong in the protocol.

The result is scientifically useful because the negative conclusion is specific. It rules out a family of tested feature maps and successor constructions under a particular public-data scenario generator, grouped development procedure, resource envelope, and comparator set. It does not rule out all QML, all cislunar machine learning, or all future quantum hardware. Such broad claims would require evidence absent here, including separate tasks, new preregistration, scalability studies, noise analysis, hardware measurements, and appropriate classical resource comparisons.

The physics-residual control is a key methodological result. A generic classical baseline is not sufficient when the prediction target contains structure that a low-fidelity model can represent directly. C06's performance shows that, for this benchmark, the available physical residual structure is more informative than the tested quantum fidelity features. That observation does not establish a theorem about quantum circuits; it establishes a task-specific comparison of inductive bias, the kinds of patterns each model is built to represent, that future proposals must beat fairly.

The retained Q02/Q03 status also matters. Many reports collapse all non-advancing paths into a single "failure" category. Here, the first-rung tasks ran correctly, but the candidates were ineligible for larger-rung computation under a rule frozen before results. Keeping `not_reached_under_frozen_eligibility` separate from a task failure makes the evidence ledger more truthful and prevents a later analyst from inventing a selection path that never existed.

The paper's societal relevance is therefore not a claim of operational benefit. It is a reusable, human-centered research pattern for high-consequence AI: include an eligible simulator-validation scope, freeze data transformations and groups, use a physics-aware control, compare against matched classical alternatives, retain null outcomes, and make the claim boundary checkable from source artifacts. This helps researchers, reviewers, and future decision-tool designers distinguish a reproducible negative result from evidence for automation. Its value is in reducing the risk that an exciting but unsupported model claim is presented as a safe decision-support capability.

The HCI implication is deliberately limited. The present results do not show that a planner would understand, trust, reject, or use the evidence package well. They specify what a later human-centered evaluation should not omit: provenance, model scope, uncertainty, failure visibility, and a non-automated fallback. Testing those requirements with intended users remains future work.

6.1 Implications for QML benchmarking

The QML literature has repeatedly shown that benchmark conclusions depend on which classical methods are made available to the comparison [1,15]. The D046 result gives a concrete trajectory application example. Comparing only with C06 would make the candidate's raw 5.80% difference look promising. The matched TWO-RBF alternative explains enough of that difference that the candidate fails the frozen 5% rule. The interpretation changes because the question changes from "is the candidate better than one baseline?" to "is its incremental mechanism still supported when a classical model is given the same residual construction?"

This contrast does not show that the matched control is the universally best classical model. It shows why a QML result needs multiple controls chosen to test the claimed mechanism. A fair comparison is not one that intentionally makes every model identical; it is one that removes alternative explanations for the purported QML contribution. In this benchmark, the alternative explanation was a classical residual representation, and it remained competitive.

6.2 Safety objective as a distinct scientific question

The FQK and CSAFE-RF records demonstrate that predictive metrics must be tied to both label meaning and intended action. In these executed analyses, the positive label is feasible. AUROC measures overall ranking, Brier score measures probability error, and recall measures how many actually feasible options are retained. Precision and the false-positive rate address the different risk of admitting an actually infeasible option. FQK remains negative under its frozen multi-metric rule even though its AUROC is above random, because its Brier score and feasible-case recall did not match the classical controls. This guards against promoting a model by whichever metric looks most favorable afterward.

The higher-recall classical result is not a rescue or a safety result. It is a future design lesson: a screening protocol should define, before fitting, a minimum feasible-case recall for usefulness, a maximum infeasible-case false-positive rate for protection, a calibration requirement, and the action taken when the model abstains. Those quantities cannot be replaced by a single "safety" score. Testing such a policy requires a separately authorized experiment; using it to alter Gate 5 after inspection would invalidate the original test.

7 Limitations and Threats to Validity

First, the study is public-data and development-only. Calibration and final-test data were not read. The paper cannot estimate final-test error, operational calibration, mission-loop behavior, propellant savings, or flight readiness. The use of Artemis-related public work establishes relevance of the problem framing, not access to mission-owned truth data or authority to draw mission conclusions.

Second, simulator agreement is bounded. The Gate 3 results support the eligible public endpoint comparisons, but they do not validate every force model, contingency, navigation product, or human-rating consideration. RTC3 is source-ineligible, and some generated scenario decisions do not have an independently feasible numerical reference. Those limitations remain in the dataset rather than being removed after model results are observed.

Third, every quantum experiment is an exact classical statevector simulation with at most eight qubits, run within a laptop-bounded resource envelope. The runs therefore omit finite measurement sampling (shot noise), imperfect gates, device-connection limits, the time required to prepare an encoded state, and transpilation, which maps an abstract circuit to a particular processor. They report neither hardware calibration nor end-to-end quantum runtime and cannot demonstrate quantum speedup, hardware efficiency, or practical quantum advantage. Any such claim would require these costs to be measured against a fair classical resource baseline.

Fourth, the methods are limited by their frozen design. The chosen preprocessing, feature maps, sample rungs, controls, and thresholds make the comparison interpretable but restrict the scope of inference. A different method can be scientifically tested only through a new protocol, not by modifying the completed campaign until a favorable result appears.

Finally, the literature synthesis is bounded, not systematic. The inclusion set provides methodological context and control requirements but does not prove that no related method exists. This is why the paper makes a benchmark claim rather than a novelty or universal-performance claim.

8 Future Research Boundary

The negative results motivate questions about task-specific symmetry, representation, and residual structure, but the completed evidence does not answer them. A future geometric-QML direction was discussed as a hypothesis, not as a result. Here a symmetry would be a transformation, such as rotating all relevant vectors together, that preserves the physical relationship between inputs and labels. An equivariant model is designed so that its output changes consistently under such a transformation. The first unanswered question is whether any nontrivial symmetry actually survives fixed Earth–Moon–Sun ephemerides, thrust constraints, and reference-frame choices. Equivariance alone would not imply a QML advantage over classical equivariant models.

A future protocol would require a mathematical contract, a symmetry audit, baseline reproduction, a representation- and parameter-matched classical comparison, ablations of symmetry and initialization, resource accounting, and explicit rejection criteria. The D054-A/P018 binding freeze records how a future audit could specify its representation, dynamical Lie algebra (DLA) basis, allowed circuit generators, development-only case groups, numerical tolerance, output paths, and figures. A DLA describes the transformations that can be generated by the allowed circuit operations. Closure decision D055-C declined to execute the audit. Accordingly, this manuscript contains no DOP853 trajectory replay (a high-order numerical integration check), no singular-value-decomposition (SVD) rank or nullspace result, no new symmetry dataset, no variational circuit training, and no geometric-QML performance claim.

This boundary is not a weakness in reporting. It prevents a promising idea from being laundered into evidence through narrative. The appropriate next step, if future authors choose to reopen research, is a separately approved protocol that can reject the hypothesis as readily as it can support it.

9 Data, Code, and Figure Availability

The source repository, governance records, machine-readable reporting outputs, figure registry, and the published source release are available at the project repository. The published release record is release v0.3.0. This paper package copies the exact figures used in the manuscript and the three core result tables plus figure registry. The package does not include calibration or final-test rows, mission-owned evidence, trained weights, or unexecuted future-protocol outputs.

10 Conclusion

Under the frozen public-data development benchmark, the tested QML candidates did not outperform strong classical controls, and no tested candidate is eligible for a Gate 6 mission experiment. The primary whole-state fidelity kernel (Q01), projected-kernel follow-up (Q01b), feasibility-only kernel (FQK), and all 16 bounded successors produced negative outcomes under their stated criteria. The strongest successor appeared better than the C06 physics-residual control, but its advantage over a like-for-like classical alternative was only 4.65%, below the 5% threshold fixed before execution.

The study's contribution is a reproducible negative result with an explicit governance and human-centered claim structure. It reports what was tested, what did not advance, what remains a future lesson, and what a human reviewer would need to see before considering further authorization. Its practical societal value is a more accountable way to evaluate AI evidence for high-consequence decision contexts, not a deployed mission capability. It does not establish quantum advantage, fuel savings, mission utility, hardware performance, flight readiness, NASA approval, or an invented QML model. That restraint is necessary for the result to be useful.

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